Supplementary Material for UNT-STLLM

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I. Paired Comparisons

The following comparisons quantify error changes under the same training protocol. Forecasting comparisons use the complete training split, whereas adaptation comparisons retain the fixed target support.

A. Differences across Seeds

Comparisons are paired within seed. For forecasting, each difference is the reference error minus the full model error. For adaptation, each difference is the independent initialization error minus the transferred initialization error. Table S1 gives the mean paired error difference with a 95% interval computed as $\bar{d} \pm t_{0.975,4} s_d/\sqrt{5}$, where s_d is the sample standard deviation of the five paired differences. These intervals summarize training variation across seeds for each comparison. The relative gain divides the difference in aggregate means by the reference mean. WAPE gains in original metric units are percentage points. The source data retain the mean of the five paired percentage gains separately.

TABLE S1. Paired improvements in forecasting error. Positive differences indicate lower error for UNT-STLLM relative to the listed reference, or for transferred relative to independent initialization. Intervals summarize seed variation over five paired runs. MAE/RMSE use flow counts, whereas WAPE differences use percentage points.

Comparison	Task or direction	Metric	Mean gain	95% interval	Gain (%)
Base	Taxi drop-off	MAE	0.1093	[0.0311, 0.1876]	2.18
Base	Taxi drop-off	RMSE	0.1793	[0.0281, 0.3305]	2.04
Base	Taxi drop-off	WAPE	0.4172	[0.1186, 0.7159]	2.18
ST-LLM+	Taxi drop-off	MAE	0.1136	[-0.0235, 0.2507]	2.26
ST-LLM+	Taxi drop-off	RMSE	0.2040	[0.0501, 0.3580]	2.32
ST-LLM+	Taxi drop-off	WAPE	0.4333	[-0.0897, 0.9563]	2.26
Base	Bicycle drop-off	MAE	0.0014	[-0.0057, 0.0085]	0.07
Base	Bicycle drop-off	RMSE	-0.0072	[-0.0268, 0.0123]	-0.26
Base	Bicycle drop-off	WAPE	0.0274	[-0.1163, 0.1711]	0.07
ST-LLM+	Bicycle drop-off	MAE	0.0232	[0.0071, 0.0392]	1.22
ST-LLM+	Bicycle drop-off	RMSE	0.0669	[0.0261, 0.1077]	2.35
ST-LLM+	Bicycle drop-off	WAPE	0.4682	[0.1441, 0.7923]	1.22
Base	Taxi pick-up	MAE	0.0968	[0.0338, 0.1597]	1.86
Base	Taxi pick-up	RMSE	0.1353	[-0.0166, 0.2872]	1.51
Base	Taxi pick-up	WAPE	0.3661	[0.1281, 0.6040]	1.86
ST-LLM+	Taxi pick-up	MAE	0.0861	[0.0407, 0.1314]	1.66
ST-LLM+	Taxi pick-up	RMSE	0.2091	[0.1066, 0.3116]	2.31
ST-LLM+	Taxi pick-up	WAPE	0.3255	[0.1541, 0.4969]	1.66
Base	Bicycle pick-up	MAE	-0.0088	[-0.0186, 0.0011]	-0.44
Base	Bicycle pick-up	RMSE	-0.0256	[-0.0492, -0.0021]	-0.84
Base	Bicycle pick-up	WAPE	-0.1766	[-0.3756, 0.0223]	-0.44
ST-LLM+	Bicycle pick-up	MAE	0.0099	[-0.0068, 0.0265]	0.49
ST-LLM+	Bicycle pick-up	RMSE	0.0173	[-0.0018, 0.0364]	0.56
ST-LLM+	Bicycle pick-up	WAPE	0.1984	[-0.1367, 0.5335]	0.49
Transferred	Taxi pick-up→drop-off	MAE	0.4576	[0.3857, 0.5295]	8.12
Transferred	Taxi pick-up→drop-off	RMSE	0.9484	[0.7670, 1.1298]	9.40
Transferred	Taxi pick-up→drop-off	WAPE	1.7462	[1.4721, 2.0203]	8.12
Transferred	Taxi drop-off→pick-up	MAE	0.4343	[0.3529, 0.5158]	7.42
Transferred	Taxi drop-off→pick-up	RMSE	0.7695	[0.3193, 1.2197]	7.49
Transferred	Taxi drop-off→pick-up	WAPE	1.6432	[1.3351, 1.9512]	7.42
Transferred	Bicycle pick-up→drop-off	MAE	0.0889	[0.0791, 0.0988]	4.42
Transferred	Bicycle pick-up→drop-off	RMSE	0.1954	[0.1721, 0.2187]	6.39
Transferred	Bicycle pick-up→drop-off	WAPE	1.7981	[1.5995, 1.9968]	4.42
Transferred	Bicycle drop-off→pick-up	MAE	0.0930	[0.0756, 0.1104]	4.37
Transferred	Bicycle drop-off→pick-up	RMSE	0.1996	[0.1674, 0.2318]	5.94
Transferred	Bicycle drop-off→pick-up	WAPE	1.8731	[1.5230, 2.2232]	4.37

II. Consistency across Environments

Table S2 compares the common configurations in environments A and B. Full model means differ by approximately 0.41% on taxi drop-off and 0.19% on bicycle drop-off. The comparison covers the combined hardware and software settings. Controlled ablations use the full model from the same environment, preserving their local reference.

TABLE S2
MAE means with sample standard deviations over five seeds for both execution environments. Percentage changes compare the environment B mean with the environment A mean. Hardware assignments appear in the main experimental setup.

Condition	Task	Environment A	Environment B	Change (%)
Full model	Taxi drop-off	4.9077 ± 0.0380	4.9279 ± 0.0592	+0.411
Full model	Bicycle drop-off	1.8793 ± 0.0036	1.8828 ± 0.0094	+0.187
Zero output, rank 8	Taxi drop-off	4.9326 ± 0.0457	4.9557 ± 0.0701	+0.469
Zero output, rank 8	Bicycle drop-off	1.8816 ± 0.0080	1.8809 ± 0.0082	-0.039
Zero output, rank 64/62	Taxi drop-off	4.9624 ± 0.0211	4.9661 ± 0.0507	+0.074
Zero output, rank 64/62	Bicycle drop-off	1.8833 ± 0.0058	1.8822 ± 0.0090	-0.056

III. Backbone Computation

The latent node features $Z \in \mathbb{R}^{N \times d}$ enter K GPT-2 blocks. Each block has m attention heads with width $d_h = d/m$. The features F^{l-1} enter block l, where $U^l = \mathrm{LN}_{1,l}(F^{l-1})$. The set $\mathcal{A}$ identifies blocks with relation biases and low rank attention updates. The attention matrices A_h^l follow the softmax with learned relation biases defined in the main method.

GPT-2 first adds pretrained positional embeddings to the continuous node features,

$$F^0 = \mathcal{D}(Z + E_{\mathrm{pos}}). \tag{S1}$$

Here $E_{\mathrm{pos}} \in \mathbb{R}^{N \times d}$ contains pretrained positional embeddings. The operator $\mathcal{D}$ applies dropout during training and is the identity during evaluation. Each block forms its combined QKV projection,

$$[Q^l \,\|\, K^l \,\|\, V^l] = U^l W_{\mathrm{qkv}}^l + b_{\mathrm{qkv}}^l. \tag{S2}$$

The combined projection is split into m heads. For $l \in \mathcal{A}$, W_{qkv}^l and W_{out}^l include the low rank updates defined in the main method. Other backbone projections retain their pretrained values. The relation term $\alpha_l B^l$ in the attention equation is zero for $l \notin \mathcal{A}$. Each head combines the value features using its attention weights. We concatenate the head outputs into O^l and project them to J^l,

$$\begin{aligned} O^l &= \Big\|_{h=1}^{m} \left[\mathcal{D}(A_h^l) V_h^l\right], \\ J^l &= O^l W_{\mathrm{out}}^l + b_{\mathrm{out}}^l. \end{aligned} \tag{S3}$$

Attention residual states G^l and normalized states T^l enter the feedforward residual,

$$\begin{aligned} G^l &= F^{l-1} + \mathcal{D}(J^l), \\ T^l &= \mathrm{LN}_{2,l}(G^l), \\ F^l &= G^l + \mathcal{D}(\mathrm{FFN}_{0,l}(T^l)). \end{aligned} \tag{S4}$$

The feedforward sublayer uses GELU activation ϕ and hidden width $d_f = 4d$. Its frozen matrices are $W_{1,l} \in \mathbb{R}^{d \times d_f}$ and $W_{2,l} \in \mathbb{R}^{d_f \times d}$. The intermediate features $C_{\mathrm{ff}}^l \in \mathbb{R}^{N \times d_f}$ and sublayer output are

$$\begin{aligned} C_{\mathrm{ff}}^l &= \phi(T^l W_{1,l} + b_{1,l}), \\ \mathrm{FFN}_{0,l}(T^l) &= C_{\mathrm{ff}}^l W_{2,l} + b_{2,l}. \end{aligned} \tag{S5}$$

Final layer normalization supplies the node features,

$$F = \mathrm{LN}_{\mathrm{final}}(F^K) \in \mathbb{R}^{N \times d}. \tag{S6}$$

IV. Additional Measurement Details

The main manuscript defines the forecasting metrics and training protocol. This section records the additional details of resource measurement used for the exact tables below.

Environment A contains the principal forecasting comparisons. Environment B contains relation controls plus adaptation experiments. Hardware assignments are given in the main experimental setup. Environment A uses PyTorch 2.8.0 with CUDA 12.8.

Inference uses the restored model selected by validation MAE after the optimizer and other training states have been released. Batch size is 20, with three warmup batches and ten timed batches per run. CUDA synchronization brackets every timed batch.

Each run contributes its median batch latency. Throughput is batch size divided by that median. Memory fields distinguish peak allocated memory from peak reserved memory. Training peaks cover the training passes, and training time per epoch includes optimizer initialization in the first epoch.

V. Forecasting Results for Each Seed

The following individual measurements support the main aggregate comparison.

TABLE S3. Forecasting scores for each seed under the matched comparison protocol. Each row averages 12 forecast steps. MAE/RMSE use flow counts, whereas WAPE is a percentage.

Task	Method	Seed	MAE	RMSE	WAPE (%)
Taxi pick-up	UNT-STLLM (base)	2026	5.148732	8.882710	19.479155
Taxi pick-up	UNT-STLLM (base)	2027	5.162608	8.886866	19.531658
Taxi pick-up	UNT-STLLM (base)	2028	5.199163	8.943802	19.669965
Taxi pick-up	UNT-STLLM (base)	2029	5.217095	9.056721	19.737597
Taxi pick-up	UNT-STLLM (base)	2030	5.250233	9.085901	19.862932
Taxi pick-up	ST-LLM+	2026	5.181208	9.053737	19.601813
Taxi pick-up	ST-LLM+	2027	5.179823	9.093773	19.596638
Taxi pick-up	ST-LLM+	2028	5.179805	8.993891	19.596484
Taxi pick-up	ST-LLM+	2029	5.203621	9.045808	19.686603
Taxi pick-up	ST-LLM+	2030	5.179883	9.037829	19.596897
Taxi pick-up	UNT-STLLM	2026	5.110692	8.880655	19.335072
Taxi pick-up	UNT-STLLM	2027	5.094800	8.808513	19.275050
Taxi pick-up	UNT-STLLM	2028	5.118076	8.853222	19.362932
Taxi pick-up	UNT-STLLM	2029	5.054315	8.735769	19.121961
Taxi pick-up	UNT-STLLM	2030	5.116165	8.901471	19.355866
Bicycle pick-up	UNT-STLLM (base)	2026	1.981836	3.049519	39.917430
Bicycle pick-up	UNT-STLLM (base)	2027	1.996853	3.082027	40.220082
Bicycle pick-up	UNT-STLLM (base)	2028	1.981480	3.051753	39.910370
Bicycle pick-up	UNT-STLLM (base)	2029	1.982308	3.056537	39.926989
Bicycle pick-up	UNT-STLLM (base)	2030	1.983501	3.054753	39.951102
Bicycle pick-up	ST-LLM+	2026	1.996763	3.107548	40.218249
Bicycle pick-up	ST-LLM+	2027	1.990383	3.073387	40.089650
Bicycle pick-up	ST-LLM+	2028	2.003951	3.108166	40.362866
Bicycle pick-up	ST-LLM+	2029	2.006164	3.103785	40.407529
Bicycle pick-up	ST-LLM+	2030	2.021821	3.116184	40.723083
Bicycle pick-up	UNT-STLLM	2026	1.997265	3.095457	40.228396
Bicycle pick-up	UNT-STLLM	2027	1.993233	3.077758	40.147070
Bicycle pick-up	UNT-STLLM	2028	1.997527	3.088697	40.233710
Bicycle pick-up	UNT-STLLM	2029	1.989650	3.082463	40.074877
Bicycle pick-up	UNT-STLLM	2030	1.992141	3.078228	40.125166
Taxi drop-off	UNT-STLLM (base)	2026	5.061980	8.885469	19.318379
Taxi drop-off	UNT-STLLM (base)	2027	4.940363	8.619628	18.854210
Taxi drop-off	UNT-STLLM (base)	2028	4.970063	8.693327	18.967254
Taxi drop-off	UNT-STLLM (base)	2029	5.045113	8.809189	19.253733
Taxi drop-off	UNT-STLLM (base)	2030	5.067667	8.881434	19.340167
Taxi drop-off	ST-LLM+	2026	4.971083	8.758606	18.971261
Taxi drop-off	ST-LLM+	2027	5.177318	8.953428	19.757833
Taxi drop-off	ST-LLM+	2028	5.018219	8.862049	19.151103
Taxi drop-off	ST-LLM+	2029	4.922446	8.641682	18.785630
Taxi drop-off	ST-LLM+	2030	5.017506	8.797007	19.148360
Taxi drop-off	UNT-STLLM	2026	4.971569	8.677397	18.973004
Taxi drop-off	UNT-STLLM	2027	4.901377	8.591601	18.705297
Taxi drop-off	UNT-STLLM	2028	4.897155	8.606251	18.689130
Taxi drop-off	UNT-STLLM	2029	4.899197	8.567719	18.696911
Taxi drop-off	UNT-STLLM	2030	4.869365	8.549600	18.583166
Bicycle drop-off	UNT-STLLM (base)	2026	1.881068	2.760893	38.027205
Bicycle drop-off	UNT-STLLM (base)	2027	1.875928	2.769277	37.923243
Bicycle drop-off	UNT-STLLM (base)	2028	1.876279	2.758269	37.930336
Bicycle drop-off	UNT-STLLM (base)	2029	1.886959	2.787989	38.146233
Bicycle drop-off	UNT-STLLM (base)	2030	1.882959	2.769662	38.065446
Bicycle drop-off	ST-LLM+	2026	1.912274	2.874945	38.658331
Bicycle drop-off	ST-LLM+	2027	1.885942	2.801003	38.125632
Bicycle drop-off	ST-LLM+	2028	1.891708	2.815274	38.242336
Bicycle drop-off	ST-LLM+	2029	1.913638	2.871687	38.685884
Bicycle drop-off	ST-LLM+	2030	1.908626	2.853892	38.584314
Bicycle drop-off	UNT-STLLM	2026	1.881108	2.783973	38.028030
Bicycle drop-off	UNT-STLLM	2027	1.877797	2.771216	37.961064
Bicycle drop-off	UNT-STLLM	2028	1.881572	2.783011	38.037472
Bicycle drop-off	UNT-STLLM	2029	1.882288	2.783795	38.051943

Task	Method	Seed	MAE	RMSE	WAPE (%)
Bicycle drop-off	UNT-STLLM	2030	1.873635	2.760278	37.876876

VI. Additional Attention Diagnostics

Table S4 reports validation diagnostics of weights after softmax from environment A. The diagnostics use models selected by validation MAE. Each run aggregates the final two relation blocks over heads and validation batches, including the padded final batch. Negative fractions are weighted by element counts. Row sum errors are weighted by row counts. Masked magnitudes are averaged over masked entries. Bidirectional conditions have no masked entries.

The table lists each unit and layer. Aggregate means and sample standard deviations over five runs are retained in the accompanying data file. The measurements concern weights after softmax. Attention with biases added before softmax preserves nonnegative weights and unit row sums before dropout by construction.

TABLE S4. Validation diagnostics of relation residuals added after softmax under the matched comparison protocol. Each row identifies one layer for one seed. Negative entries are reported as a percentage. Row sum error is averaged over rows. Masked absolute weights are averaged over masked entries. Bidirectional attention has no masked entries.

Condition	Task	Seed	Layer	Negative (%)	Row sum error	Masked mean $\lvert A\rvert$
After, bidirectional	Bicycle drop-off	2026	3	42.73	5.039	N/A
After, bidirectional	Bicycle drop-off	2026	4	50.99	6.401	N/A
After, bidirectional	Bicycle drop-off	2027	3	62.55	3.878	N/A
After, bidirectional	Bicycle drop-off	2027	4	30.21	5.558	N/A
After, bidirectional	Bicycle drop-off	2028	3	34.05	3.980	N/A
After, bidirectional	Bicycle drop-off	2028	4	40.42	5.265	N/A
After, bidirectional	Bicycle drop-off	2029	3	31.04	5.472	N/A
After, bidirectional	Bicycle drop-off	2029	4	52.50	5.211	N/A
After, bidirectional	Bicycle drop-off	2030	3	53.78	4.157	N/A
After, bidirectional	Bicycle drop-off	2030	4	61.30	6.168	N/A
After, bidirectional	Taxi drop-off	2026	3	73.00	5.210	N/A
After, bidirectional	Taxi drop-off	2026	4	67.60	9.214	N/A
After, bidirectional	Taxi drop-off	2027	3	62.60	5.052	N/A
After, bidirectional	Taxi drop-off	2027	4	65.33	5.903	N/A
After, bidirectional	Taxi drop-off	2028	3	74.59	3.835	N/A
After, bidirectional	Taxi drop-off	2028	4	73.89	7.471	N/A
After, bidirectional	Taxi drop-off	2029	3	69.30	4.512	N/A
After, bidirectional	Taxi drop-off	2029	4	65.97	8.641	N/A
After, bidirectional	Taxi drop-off	2030	3	80.54	7.241	N/A
After, bidirectional	Taxi drop-off	2030	4	58.11	5.520	N/A
After, causal	Bicycle drop-off	2026	3	27.53	9.319	0.0405
After, causal	Bicycle drop-off	2026	4	59.87	8.183	0.0542
After, causal	Bicycle drop-off	2027	3	41.26	4.769	0.0277
After, causal	Bicycle drop-off	2027	4	72.18	23.001	0.1036
After, causal	Bicycle drop-off	2028	3	55.96	6.814	0.0463
After, causal	Bicycle drop-off	2028	4	39.06	6.504	0.0456
After, causal	Bicycle drop-off	2029	3	34.53	4.451	0.0185
After, causal	Bicycle drop-off	2029	4	47.41	7.753	0.0520
After, causal	Bicycle drop-off	2030	3	60.39	7.210	0.0442
After, causal	Bicycle drop-off	2030	4	48.78	7.844	0.0396
After, causal	Taxi drop-off	2026	3	80.89	7.741	0.0345
After, causal	Taxi drop-off	2026	4	68.90	10.192	0.0452
After, causal	Taxi drop-off	2027	3	59.34	4.172	0.0175
After, causal	Taxi drop-off	2027	4	61.53	6.270	0.0331
After, causal	Taxi drop-off	2028	3	83.45	6.179	0.0255
After, causal	Taxi drop-off	2028	4	66.80	7.154	0.0323
After, causal	Taxi drop-off	2029	3	59.71	4.353	0.0169
After, causal	Taxi drop-off	2029	4	69.10	9.349	0.0425
After, causal	Taxi drop-off	2030	3	74.93	6.696	0.0283
After, causal	Taxi drop-off	2030	4	71.17	10.327	0.0438

The main manuscript presents the taxi pick-up relation maps. Fig. S1 provides the additional taxi drop-off case using the comparison with the same block input.

VII. Controlled Comparisons for Each Seed

Table S5 lists all completed runs in the controlled comparisons. Each environment has its own full model reference. The environment A full model runs are listed in Table S3. The remaining runs are listed here. Updated counts include parameters with gradients rather than unused word embeddings.

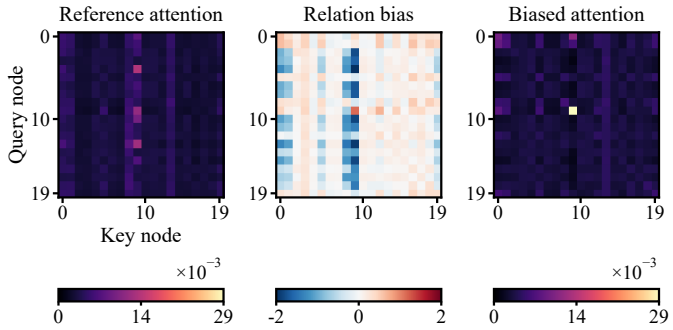


Fig. S1. Attention in block 5 and learned relation biases for taxi drop-off at a fixed block input. The displayed nodes are 0–19 from 266 nodes, with normalization over all nodes and scaled bias rows centered for display. The outer color bars give attention weights in units of 10^{-3}, and the middle bar gives bias values.

TABLE S5. Errors and effective update counts for each seed in the controlled conditions. A and B denote the main comparison environment and the transfer environment. MAE and RMSE use flow counts, WAPE is a percentage, and updated counts are in millions.

Environment	Task	Code	Seed	MAE	RMSE	WAPE (%)	Updated (M)
B	bike_drop	Cap	2026	1.892671	2.807986	38.261737	2.765837
B	bike_drop	Cap	2027	1.874716	2.767822	37.898805	2.765837
B	bike_drop	Cap	2028	1.891518	2.808448	38.238464	2.765837
B	bike_drop	Cap	2029	1.876345	2.770717	37.931725	2.765837
B	bike_drop	Cap	2030	1.875812	2.767893	37.920882	2.765837
B	bike_drop	Dyn	2026	1.894914	2.815993	38.307082	2.637199
B	bike_drop	Dyn	2027	1.876919	2.769070	37.943235	2.637199
B	bike_drop	Dyn	2028	1.876490	2.762302	37.934599	2.637199
B	bike_drop	Dyn	2029	1.885582	2.796180	38.118520	2.637199
B	bike_drop	Dyn	2030	1.888058	2.802979	38.168437	2.637199
B	bike_drop	M	2026	1.875899	2.765794	37.922720	2.762199
B	bike_drop	M	2027	1.876405	2.769246	37.932888	2.762199
B	bike_drop	M	2028	1.876085	2.762010	37.926402	2.762199
B	bike_drop	M	2029	1.889423	2.799515	38.196196	2.762199
B	bike_drop	M	2030	1.896123	2.810280	38.331531	2.762199
B	bike_drop	NR	2026	1.894009	2.806525	38.288887	2.762199
B	bike_drop	NR	2027	1.872314	2.763025	37.850200	2.762199
B	bike_drop	NR	2028	1.876701	2.764559	37.938888	2.762199
B	bike_drop	NR	2029	1.879028	2.777576	37.986049	2.762199
B	bike_drop	NR	2030	1.882301	2.777412	38.052158	2.762199
B	bike_drop	Stat0	2026	1.880120	2.782249	38.008153	2.393175
B	bike_drop	Stat0	2027	1.875962	2.770442	37.923995	2.393175
B	bike_drop	Stat0	2028	1.879848	2.772840	38.002478	2.393175
B	bike_drop	Stat0	2029	1.885463	2.779805	38.116121	2.393175
B	bike_drop	Stat0	2030	1.876053	2.768372	37.925798	2.393175
B	taxi_drop	Cap	2026	4.999019	8.711228	19.077866	2.788365
B	taxi_drop	Cap	2027	4.915140	8.632659	18.757880	2.788365
B	taxi_drop	Cap	2028	4.908375	8.578708	18.732016	2.788365
B	taxi_drop	Cap	2029	4.990233	8.682026	19.044302	2.788365
B	taxi_drop	Cap	2030	5.017810	8.814120	19.149642	2.788365
B	taxi_drop	Dyn	2026	4.949531	8.591357	18.889006	2.641295
B	taxi_drop	Dyn	2027	4.895206	8.596799	18.681677	2.641295
B	taxi_drop	Dyn	2028	4.929335	8.611439	18.811950	2.641295
B	taxi_drop	Dyn	2029	4.924763	8.664162	18.794474	2.641295
B	taxi_drop	Dyn	2030	4.939266	8.625742	18.849897	2.641295
B	taxi_drop	M	2026	4.919613	8.622517	18.774967	2.782807
B	taxi_drop	M	2027	4.911200	8.560765	18.742927	2.782807
B	taxi_drop	M	2028	4.890676	8.600420	18.664353	2.782807
B	taxi_drop	M	2029	4.887176	8.596942	18.651157	2.782807
B	taxi_drop	M	2030	5.030877	8.769922	19.199406	2.782807
B	taxi_drop	NR	2026	5.051363	8.727664	19.277537	2.782807
B	taxi_drop	NR	2027	4.919263	8.670360	18.773603	2.782807
B	taxi_drop	NR	2028	4.885665	8.562183	18.645234	2.782807
B	taxi_drop	NR	2029	4.915197	8.632168	18.757983	2.782807
B	taxi_drop	NR	2030	5.007053	8.755594	19.108451	2.782807
B	taxi_drop	Stat0	2026	4.974559	8.682197	18.984700	2.413783
B	taxi_drop	Stat0	2027	4.908395	8.630221	18.732134	2.413783
B	taxi_drop	Stat0	2028	4.916817	8.608074	18.764201	2.413783
B	taxi_drop	Stat0	2029	4.916871	8.613800	18.764316	2.413783
B	taxi_drop	Stat0	2030	4.896386	8.589253	18.686342	2.413783

Environment	Task	Code	Seed	MAE	RMSE	WAPE (%)	Updated (M)
A	bike_drop	2L	2026	1.899245	2.831644	38.394687	17.652183
A	bike_drop	2L	2027	1.892031	2.819469	38.248824	17.652183
A	bike_drop	2L	2028	1.883839	2.796536	38.083215	17.652183
A	bike_drop	2L	2029	1.883214	2.792272	38.070500	17.652183
A	bike_drop	2L	2030	1.883308	2.795218	38.072346	17.652183
A	bike_drop	NR	2026	1.891601	2.798116	38.240153	2.762199
A	bike_drop	NR	2027	1.871621	2.757662	37.836196	2.762199
A	bike_drop	NR	2028	1.884409	2.790998	38.094691	2.762199
A	bike_drop	NR	2029	1.884809	2.786851	38.102852	2.762199
A	bike_drop	NR	2030	1.875545	2.766721	37.915536	2.762199
A	bike_drop	NR-T	2026	1.881585	2.783288	38.037675	2.688471
A	bike_drop	NR-T	2027	1.878795	2.775420	37.981229	2.688471
A	bike_drop	NR-T	2028	1.891187	2.807718	38.231730	2.688471
A	bike_drop	NR-T	2029	1.876739	2.767280	37.939680	2.688471
A	bike_drop	NR-T	2030	1.876832	2.767695	37.941477	2.688471
A	bike_drop	Post-C	2026	1.881994	2.782926	38.045936	2.762199
A	bike_drop	Post-C	2027	1.880748	2.785426	38.020710	2.762199
A	bike_drop	Post-C	2028	1.897907	2.815652	38.367632	2.762199
A	bike_drop	Post-C	2029	1.881508	2.793093	38.035961	2.762199
A	bike_drop	Post-C	2030	1.883256	2.786054	38.071392	2.762199
A	bike_drop	Rnd	2026	1.886249	2.794722	38.131899	2.762199
A	bike_drop	Rnd	2027	1.889478	2.794970	38.197221	2.762199
A	bike_drop	Rnd	2028	1.885021	2.796787	38.107177	2.762199
A	bike_drop	Rnd	2029	1.886226	2.798448	38.131418	2.762199
A	bike_drop	Rnd	2030	1.875971	2.762332	37.924108	2.762199
A	bike_drop	Pre-C	2026	1.892068	2.809735	38.249666	2.762199
A	bike_drop	Pre-C	2027	1.879169	2.778118	37.988824	2.762199
A	bike_drop	Pre-C	2028	1.889592	2.794943	38.199473	2.762199
A	bike_drop	Pre-C	2029	1.888193	2.790572	38.171294	2.762199
A	bike_drop	Pre-C	2030	1.875409	2.762211	37.912840	2.762199
A	bike_drop	T	2026	1.881724	2.782366	38.040420	2.688471
A	bike_drop	T	2027	1.884045	2.784864	38.087341	2.688471
A	bike_drop	T	2028	1.886791	2.802752	38.142907	2.688471
A	bike_drop	T	2029	1.883845	2.796321	38.083255	2.688471
A	bike_drop	T	2030	1.874151	2.762256	37.887289	2.688471
A	taxi_drop	2L	2026	5.034104	8.784754	19.211759	17.672791
A	taxi_drop	2L	2027	4.900042	8.527769	18.700244	17.672791
A	taxi_drop	2L	2028	4.953449	8.603122	18.903859	17.672791
A	taxi_drop	2L	2029	4.955423	8.676362	18.911378	17.672791
A	taxi_drop	2L	2030	5.010648	8.757470	19.122436	17.672791
A	taxi_drop	NR	2026	5.008277	8.746436	19.113230	2.782807
A	taxi_drop	NR	2027	4.909294	8.581949	18.735527	2.782807
A	taxi_drop	NR	2028	4.926837	8.648956	18.802369	2.782807
A	taxi_drop	NR	2029	4.930906	8.644223	18.817942	2.782807
A	taxi_drop	NR	2030	4.887544	8.598436	18.652532	2.782807
A	taxi_drop	NR-T	2026	5.102794	8.808414	19.473850	2.709079
A	taxi_drop	NR-T	2027	5.218807	9.040031	19.916496	2.709079
A	taxi_drop	NR-T	2028	5.116836	8.914312	19.527279	2.709079
A	taxi_drop	NR-T	2029	5.137733	8.961236	19.607509	2.709079
A	taxi_drop	NR-T	2030	5.061536	8.806196	19.316989	2.709079
A	taxi_drop	Post-B	2026	4.985679	8.686660	19.026966	2.782807
A	taxi_drop	Post-B	2027	4.908034	8.607310	18.730804	2.782807
A	taxi_drop	Post-B	2028	4.888952	8.582326	18.657980	2.782807
A	taxi_drop	Post-B	2029	4.838742	8.504447	18.466382	2.782807
A	taxi_drop	Post-B	2030	4.928003	8.590331	18.807095	2.782807
A	taxi_drop	Post-C	2026	4.927080	8.595937	18.803330	2.782807
A	taxi_drop	Post-C	2027	4.890990	8.571457	18.665620	2.782807
A	taxi_drop	Post-C	2028	4.931707	8.699994	18.820970	2.782807
A	taxi_drop	Post-C	2029	4.968574	8.724342	18.961771	2.782807
A	taxi_drop	Post-C	2030	4.982095	8.664069	19.013306	2.782807
A	taxi_drop	Rnd	2026	5.010306	8.764444	19.120927	2.782807
A	taxi_drop	Rnd	2027	4.975946	8.697236	18.990140	2.782807
A	taxi_drop	Rnd	2028	4.969636	8.669648	18.965623	2.782807
A	taxi_drop	Rnd	2029	4.884598	8.521845	18.641229	2.782807
A	taxi_drop	Rnd	2030	4.894567	8.650213	18.679531	2.782807
A	taxi_drop	Pre-C	2026	4.921746	8.586002	18.782886	2.782807
A	taxi_drop	Pre-C	2027	4.924710	8.598918	18.794381	2.782807
A	taxi_drop	Pre-C	2028	4.903512	8.589801	18.713318	2.782807
A	taxi_drop	Pre-C	2029	5.041130	8.801638	19.238664	2.782807
A	taxi_drop	Pre-C	2030	4.861976	8.549266	18.554925	2.782807
A	taxi_drop	T	2026	5.119644	8.870984	19.538322	2.709079
A	taxi_drop	T	2027	5.191011	9.000799	19.810429	2.709079
A	taxi_drop	T	2028	5.077113	8.810875	19.375723	2.709079
A	taxi_drop	T	2029	5.137350	8.921457	19.606048	2.709079
A	taxi_drop	T	2030	4.936288	8.638495	18.838443	2.709079

Environment	Task	Code	Seed	MAE	RMSE	WAPE (%)	Updated (M)
A	bike_drop	Cap	2026	1.885997	2.786912	38.126896	2.765837
A	bike_drop	Cap	2027	1.885484	2.785863	38.116477	2.765837
A	bike_drop	Cap	2028	1.889644	2.799848	38.200540	2.765837
A	bike_drop	Cap	2029	1.880619	2.772890	38.018128	2.765837
A	bike_drop	Cap	2030	1.874568	2.761633	37.895756	2.765837
A	taxi_drop	Cap	2026	4.965460	8.592699	18.949755	2.788365
A	taxi_drop	Cap	2027	4.979445	8.704687	19.003077	2.788365
A	taxi_drop	Cap	2028	4.926853	8.708539	18.802569	2.788365
A	taxi_drop	Cap	2029	4.963411	8.661511	18.942031	2.788365
A	taxi_drop	Cap	2030	4.977060	8.741559	18.994209	2.788365

Codes are M for the full model, NR for zero relations, T for updates without LoRA, NR-T for both changes, Pre-C for causal attention with biases before softmax, Post-C and Post-B for biases after softmax with causal and bidirectional access, Rnd for random weights, 2L for two fully trained blocks, Cap for budget reallocation, Dyn for dynamic relations, and Stat0 for static relations initialized at zero.

VIII. Reference Analyses under Original Settings

Earlier attention and representation analyses provide design context for the main comparisons.

A. Earlier Attention Comparisons

The cumulative attention comparisons used five seeds, batch 20 and 250 epochs under the earlier protocol. The intermediate configurations separate changes in attention placement, node access and backbone update policy under those settings.

Fig. S2 compares the base with three cumulative changes. The first moves the relation term before softmax while retaining causal attention over the node sequence. The second also enables bidirectional node attention, and UNT-STLLM additionally uses LoRA for backbone adaptation. UNT-STLLM lowers average errors on both taxi tasks relative to the retrained base.

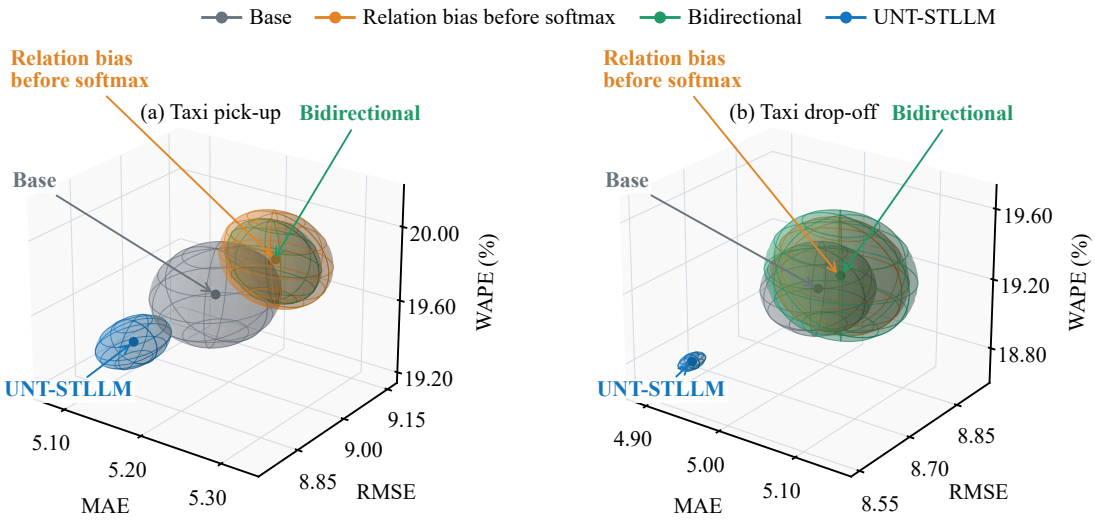


Fig. S2. MAE, RMSE and WAPE profiles under the earlier protocol for the two taxi tasks. Ellipsoid centers give means over five seeds, and semiaxes give sample standard deviations along the corresponding metric axes.

B. Earlier Inference Measurements

The following reference measurements retain their original hardware settings. The main efficiency figure reports the current comparison on the same host.

The bicycle inference measurements retain their respective experimental settings (Fig. S3). Reference MAE values come from Table III of Liu et al. [1]. Memory values follow their Fig. 4, with FPS digitized to the nearest ten. Reference measurements retain their original environment. FPS denotes frames per second. For the UNT measurements, one frame is a complete 12-step input window across all nodes. We measured prediction speed under the earlier protocol with batches of 20 windows. We recorded PyTorch peak allocated inference memory. UNT points are means over five runs, and numeric labels give GPU memory in GiB.

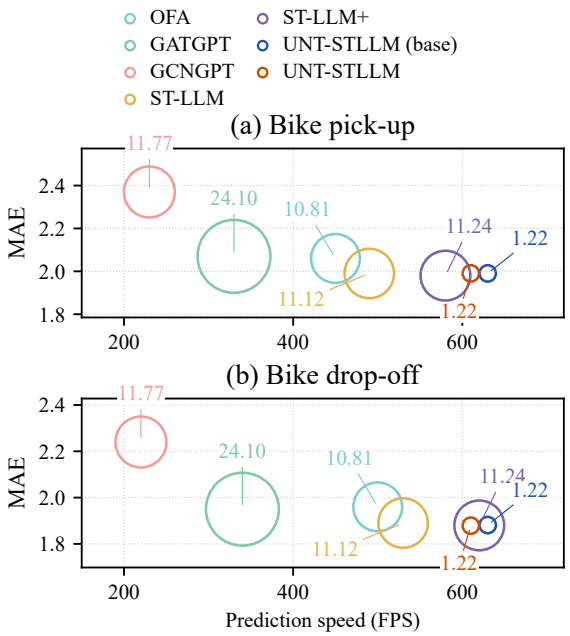


Fig. S3. CHBike inference profiles under the original measurement settings. Reference MAE and memory follow ST-LLM+ under the published measurement settings, with FPS digitized to the nearest ten, whereas UNT points are means over five seeds under the earlier protocol with batch 20. One UNT frame contains a complete 12-step input across all nodes. Hollow circle areas encode peak allocated inference memory in GiB, with exact values printed beside them.

C. Representation Analysis Settings

Representation analysis used the original UNT-STLLM evaluation configuration under the earlier protocol, with models selected by validation MAE during 250 epochs of training on the complete training dataset. The seeds were 2028 for taxi pick-up, 2027 for taxi drop-off, 2029 for bicycle pick-up and 2027 for bicycle drop-off. Attention and fusion maps used all 870 validation windows per task. Attention was averaged across heads. Fusion was grouped by the final observed hour. KL profiles averaged windows and nodes. PCA and latent traversal used all nodes in eight evenly spaced validation windows. Traversals perturbed dimensions 0, 191, 383, 575 and 767 by $0, \pm\sigma, \pm 2\sigma$, where σ is each node's posterior standard deviation. Other coordinates remained at their posterior means. Responses averaged absolute forecast changes across windows, nodes and forecast steps.

IX. Adaptation and Conditions on Target Data

The main manuscript reports adaptation plus prediction without target parameter updates. This section retains additional implementation details and earlier evaluation settings.

A. Adaptation with Fixed Target Data

The current comparison of four transfer directions uses PyTorch 2.8.0+cu128. The two UNT-STLLM initializations share the same model, training, and utility implementation. Each direction uses the complete source training split and 260 fixed target training windows per seed, with the scaler fitted to the selected target inputs. The initializations share support indices, fresh initialization of node parameters, random states, and batch order within seed. Both start with an empty optimizer state, and source test data are excluded from selection of the source model. The earlier base comparison retains its earlier environment and PyTorch 2.5.1+cu121 settings.

Each stage runs for 250 epochs with batch 20 and learning rate 3×10^{-3}. Validation MAE selects the state within each stage. The target stages each perform 3,250 optimizer updates. Transferred initialization copies the source state selected by validation MAE except for target node embeddings and static relation matrices. Independent initialization starts from the backbone pretrained on text with freshly initialized forecasting modules. UNT-STLLM uses LoRA with rank 8 in both conditions. Base retains its original update policy. Table S7 gives the earlier target metrics. Table S6 supplies the complete current results for all three metrics and four directions underlying the main transfer figure. Table S8 gives the current values for each seed. Table S9 gives the earlier values for each seed.

TABLE S6
ADAPTATION IN FOUR DIRECTIONS UNDER FIXED 10% TARGET SUPPORT

Direction	Initialization	MAE	RMSE	WAPE (%)
Taxi pick-up → drop-off	Independent	$\underline{5.633} \pm 0.044$	$\underline{10.093} \pm 0.058$	$\underline{21.50} \pm 0.17$
Taxi pick-up → drop-off	Transferred	$\mathbf{5.176} \pm 0.067$	$\mathbf{9.145} \pm 0.108$	$\mathbf{19.75} \pm 0.26$
Taxi drop-off → pick-up	Independent	$\underline{5.850} \pm 0.086$	$\underline{10.270} \pm 0.349$	$\underline{22.13} \pm 0.33$
Taxi drop-off → pick-up	Transferred	$\mathbf{5.416} \pm 0.042$	$\mathbf{9.500} \pm 0.162$	$\mathbf{20.49} \pm 0.16$
Bicycle pick-up → drop-off	Independent	$\underline{2.013} \pm 0.005$	$\underline{3.057} \pm 0.014$	$\underline{40.69} \pm 0.10$
Bicycle pick-up → drop-off	Transferred	$\mathbf{1.924} \pm 0.010$	$\mathbf{2.862} \pm 0.023$	$\mathbf{38.89} \pm 0.20$
Bicycle drop-off → pick-up	Independent	$\underline{2.127} \pm 0.012$	$\underline{3.360} \pm 0.021$	$\underline{42.84} \pm 0.24$
Bicycle drop-off → pick-up	Transferred	$\mathbf{2.034} \pm 0.005$	$\mathbf{3.160} \pm 0.011$	$\mathbf{40.96} \pm 0.10$

Entries are mean ± sample standard deviation over five seeds with the same fixed 260 target windows within seed, with lower values preferred, bold indicating the best and underlining the second best within each direction.

TABLE S7
EARLIER TARGET ADAPTATION WITH FIXED TRAINING DATA

Earlier comparison with fixed target data (original protocol)					
Target	Model	Initialization	MAE	RMSE	WAPE (%)
Taxi drop-off	UNT-STLLM (base)	Independent	5.651 ± 0.103	10.068 ± 0.191	21.57 ± 0.40
Taxi drop-off	UNT-STLLM (base)	Transferred	$\underline{5.229} \pm 0.058$	$\underline{9.237} \pm 0.121$	$\underline{19.96} \pm 0.22$
Taxi drop-off	UNT-STLLM	Independent	5.586 ± 0.084	9.956 ± 0.225	21.32 ± 0.32
Taxi drop-off	UNT-STLLM	Transferred	$\mathbf{5.180} \pm 0.051$	$\mathbf{9.129} \pm 0.107$	$\mathbf{19.77} \pm 0.19$
Bicycle drop-off	UNT-STLLM (base)	Independent	2.007 ± 0.019	3.045 ± 0.051	40.57 ± 0.38
Bicycle drop-off	UNT-STLLM (base)	Transferred	$\mathbf{1.920} \pm 0.006$	$\mathbf{2.843} \pm 0.017$	$\mathbf{38.81} \pm 0.12$
Bicycle drop-off	UNT-STLLM	Independent	2.018 ± 0.014	3.065 ± 0.030	40.80 ± 0.29
Bicycle drop-off	UNT-STLLM	Transferred	$\underline{1.924} \pm 0.011$	$\underline{2.861} \pm 0.025$	$\underline{38.90} \pm 0.22$

Entries use five seeds under the earlier protocol with fixed 10% target support, with lower errors preferred, the best result in bold and the second best underlined within each comparison.

TABLE S8. Adaptation results for each seed in four transfer directions with fixed 10% target support. Both initializations use the same 260 target windows within seed. Their scalers are identical. MAE/RMSE use flow counts, whereas WAPE is a percentage.

Direction	Initialization	Seed	MAE	RMSE	WAPE (%)
Taxi pick-up to drop-off	Independent	2026	5.680281	10.028755	21.677832
Taxi pick-up to drop-off	Independent	2027	5.623486	10.086905	21.460941
Taxi pick-up to drop-off	Independent	2028	5.584276	10.072025	21.311046
Taxi pick-up to drop-off	Independent	2029	5.600806	10.089145	21.374221
Taxi pick-up to drop-off	Independent	2030	5.678101	10.188208	21.668977
Taxi pick-up to drop-off	Transferred	2026	5.279174	9.328616	20.146903
Taxi pick-up to drop-off	Transferred	2027	5.112595	9.048052	19.511379
Taxi pick-up to drop-off	Transferred	2028	5.127509	9.132197	19.568304
Taxi pick-up to drop-off	Transferred	2029	5.201879	9.085859	19.851856
Taxi pick-up to drop-off	Transferred	2030	5.157718	9.128232	19.683741
Taxi drop-off to pick-up	Independent	2026	5.801304	10.101579	21.947867
Taxi drop-off to pick-up	Independent	2027	5.794901	9.980261	21.923784
Taxi drop-off to pick-up	Independent	2028	5.845587	10.183809	22.114991
Taxi drop-off to pick-up	Independent	2029	5.807482	10.208402	21.971353
Taxi drop-off to pick-up	Independent	2030	5.999899	10.874146	22.699103
Taxi drop-off to pick-up	Transferred	2026	5.405936	9.406794	20.452090
Taxi drop-off to pick-up	Transferred	2027	5.373418	9.439781	20.329142
Taxi drop-off to pick-up	Transferred	2028	5.465535	9.768867	20.677364
Taxi drop-off to pick-up	Transferred	2029	5.379047	9.360280	20.350275
Taxi drop-off to pick-up	Transferred	2030	5.453594	9.525096	20.632459
Bicycle pick-up to drop-off	Independent	2026	2.007266	3.032767	40.579017
Bicycle pick-up to drop-off	Independent	2027	2.015207	3.070175	40.739315
Bicycle pick-up to drop-off	Independent	2028	2.019749	3.063589	40.831505
Bicycle pick-up to drop-off	Independent	2029	2.011326	3.062575	40.661218
Bicycle pick-up to drop-off	Independent	2030	2.011006	3.058127	40.653864
Bicycle pick-up to drop-off	Transferred	2026	1.919365	2.847265	38.801550
Bicycle pick-up to drop-off	Transferred	2027	1.927729	2.880790	38.970702
Bicycle pick-up to drop-off	Transferred	2028	1.933900	2.878757	39.095624
Bicycle pick-up to drop-off	Transferred	2029	1.930169	2.873990	39.020096
Bicycle pick-up to drop-off	Transferred	2030	1.908737	2.829369	38.586323
Bicycle drop-off to pick-up	Independent	2026	2.130870	3.366056	42.920093
Bicycle drop-off to pick-up	Independent	2027	2.123382	3.348033	42.768960
Bicycle drop-off to pick-up	Independent	2028	2.135247	3.377777	43.008108

Direction	Initialization	Seed	MAE	RMSE	WAPE (%)
Bicycle drop-off to pick-up	Independent	2029	2.136725	3.378075	43.037760
Bicycle drop-off to pick-up	Independent	2030	2.107094	3.329671	42.440584
Bicycle drop-off to pick-up	Transferred	2026	2.030471	3.154041	40.897296
Bicycle drop-off to pick-up	Transferred	2027	2.026579	3.148217	40.818940
Bicycle drop-off to pick-up	Transferred	2028	2.037531	3.168841	41.039515
Bicycle drop-off to pick-up	Transferred	2029	2.034958	3.156247	40.987639
Bicycle drop-off to pick-up	Transferred	2030	2.038891	3.174214	41.066682

TABLE S9. Target adaptation results for each seed under the earlier protocol with fixed 10% target support. Initializations share target windows and normalization within seed. MAE/RMSE use flow counts, whereas WAPE is a percentage.

Target	Model	Initialization	Seed	MAE	RMSE	WAPE (%)
Bicycle drop-off	UNT-STLLM	Transferred	2026	1.939182	2.888311	39.202221
Bicycle drop-off	UNT-STLLM	Transferred	2027	1.924961	2.867617	38.914852
Bicycle drop-off	UNT-STLLM	Transferred	2028	1.923679	2.854630	38.888925
Bicycle drop-off	UNT-STLLM	Transferred	2029	1.925801	2.875052	38.931700
Bicycle drop-off	UNT-STLLM	Transferred	2030	1.908397	2.821851	38.579584
Bicycle drop-off	UNT-STLLM	Independent	2026	2.019512	3.064021	40.826610
Bicycle drop-off	UNT-STLLM	Independent	2027	2.029719	3.091809	41.032868
Bicycle drop-off	UNT-STLLM	Independent	2028	2.033315	3.099626	41.105625
Bicycle drop-off	UNT-STLLM	Independent	2029	2.000254	3.033390	40.437285
Bicycle drop-off	UNT-STLLM	Independent	2030	2.007183	3.038215	40.576585
Bicycle drop-off	Base	Transferred	2026	1.914921	2.833069	38.711750
Bicycle drop-off	Base	Transferred	2027	1.923340	2.848822	38.881990
Bicycle drop-off	Base	Transferred	2028	1.925946	2.860223	38.934856
Bicycle drop-off	Base	Transferred	2029	1.923083	2.853626	38.876824
Bicycle drop-off	Base	Transferred	2030	1.912443	2.818047	38.661359
Bicycle drop-off	Base	Independent	2026	1.981735	2.978348	40.062746
Bicycle drop-off	Base	Independent	2027	1.994393	3.011515	40.318561
Bicycle drop-off	Base	Independent	2028	2.009181	3.052253	40.617771
Bicycle drop-off	Base	Independent	2029	2.025654	3.107093	40.950349
Bicycle drop-off	Base	Independent	2030	2.024239	3.074759	40.921474
Taxi drop-off	UNT-STLLM	Transferred	2026	5.112461	8.965569	19.511192
Taxi drop-off	UNT-STLLM	Transferred	2027	5.211824	9.219558	19.889983
Taxi drop-off	UNT-STLLM	Transferred	2028	5.208202	9.222535	19.876036
Taxi drop-off	UNT-STLLM	Transferred	2029	5.227387	9.152342	19.949448
Taxi drop-off	UNT-STLLM	Transferred	2030	5.138095	9.085643	19.608575
Taxi drop-off	UNT-STLLM	Independent	2026	5.668917	10.105708	21.634288
Taxi drop-off	UNT-STLLM	Independent	2027	5.479971	9.639239	20.913208
Taxi drop-off	UNT-STLLM	Independent	2028	5.569232	9.938171	21.253895
Taxi drop-off	UNT-STLLM	Independent	2029	5.671477	10.226353	21.644150
Taxi drop-off	UNT-STLLM	Independent	2030	5.538700	9.872775	21.137244
Taxi drop-off	Base	Transferred	2026	5.256638	9.282575	20.061080
Taxi drop-off	Base	Transferred	2027	5.244404	9.270120	20.014253
Taxi drop-off	Base	Transferred	2028	5.214623	9.290705	19.900430
Taxi drop-off	Base	Transferred	2029	5.292384	9.319295	20.197209
Taxi drop-off	Base	Transferred	2030	5.139336	9.023153	19.613444
Taxi drop-off	Base	Independent	2026	5.715259	10.277471	21.811317
Taxi drop-off	Base	Independent	2027	5.586534	9.939920	21.319712
Taxi drop-off	Base	Independent	2028	5.569228	10.029860	21.253510
Taxi drop-off	Base	Independent	2029	5.580368	9.843128	21.296088
Taxi drop-off	Base	Independent	2030	5.802176	10.250900	22.142218

B. Batch Sampling at Each Epoch

For few-shot prediction, we trained UNT-STLLM for 250 epochs using 10% of training batches redrawn at each epoch. The scaler uses the full training inputs under this earlier protocol. Table S10 shows competitive forecasts across the four tasks, with the lowest errors on taxi drop-off. Bicycle forecasts are closely comparable to those of base.

X. Additional Latent Responses

Fig. S4 pairs KL profiles for latent coordinates with forecast responses to fixed perturbations around posterior means. Five fixed coordinates are shifted by zero, one, or two posterior standard deviations in each direction while all others remain fixed. Responses average absolute forecast changes over windows, nodes, and forecast steps. The profiles characterize how regularized latent coordinates participate in point forecasting without interpreting their variances as calibrated prediction intervals.

TABLE S10
FEW-SHOT PREDICTION WITH 10% TRAINING BATCHES PER EPOCH

Method	NYCTaxi Pick-up			NYCTaxi Drop-off			CHBike Pick-up			CHBike Drop-off		
	MAE	RMSE	WAPE	MAE	RMSE	WAPE	MAE	RMSE	WAPE	MAE	RMSE	WAPE
OFA	6.49	12.12	24.54%	6.27	12.10	23.92%	2.20	3.59	44.40%	2.06	3.17	41.63%
GATGPT	7.02	13.09	26.54%	6.84	13.27	26.09%	2.59	4.41	52.20%	2.50	4.07	50.64%
GCNGPT	10.31	18.82	39.02%	9.25	19.50	35.28%	2.73	4.44	55.20%	2.79	4.65	56.28%
Llama 2	5.81	10.16	21.99%	5.59	9.90	21.35%	2.24	3.58	45.20%	2.11	3.23	42.75%
ST-LLM	5.40	9.63	20.45%	5.54	9.84	21.14%	2.07	3.23	41.85%	1.93	2.88	39.21%
ST-LLM+	**5.37**	**9.31**	**19.86%**	5.46	9.77	21.12%	2.03	3.16	40.48%	1.91	2.83	39.13%
UNT-STLLM (base)	5.61	9.91	21.24%	5.45	9.72	20.83%	**2.00**	**3.11**	**40.47%**	**1.89**	**2.79**	**38.22%**
UNT-STLLM	5.66	9.87	21.40%	**5.38**	**9.53**	**20.52%**	2.02	3.14	40.67%	1.91	2.84	38.60%

Lower values are better, with the best result in bold and the second best underlined.

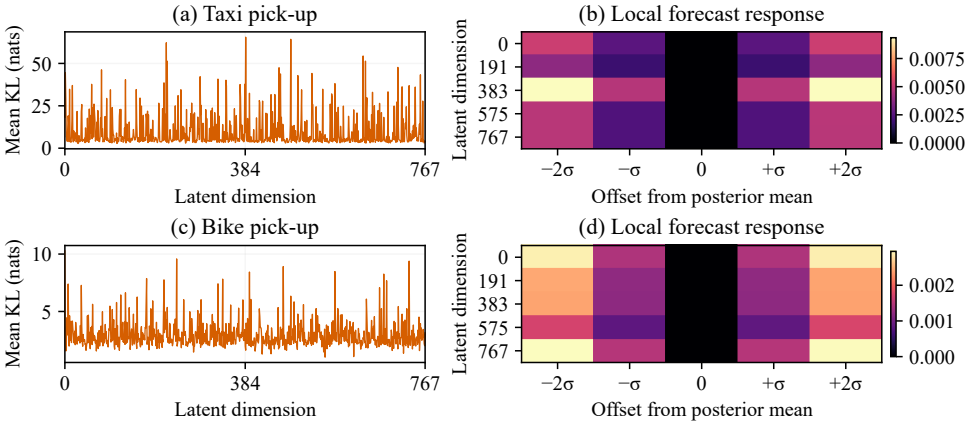


Fig. S4. Latent KL profiles and forecast responses for the taxi and bicycle pick-up tasks. Left panels average KL over all validation windows and nodes. Right panels give absolute forecast changes across eight fixed windows after five coordinates were shifted by zero, one, or two posterior standard deviations while all other coordinates remained fixed.

REFERENCES

[1] C. Liu *et al.*, "ST-LLM+: Graph enhanced spatio-temporal large language models for traffic prediction," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 8, pp. 4846–4859, 2025, doi 10.1109/TKDE.2025.3570705.